Polarimetric Analysis of Dust Continuum in Molecular Clouds

BISTRO Data Pipeline

7 апреля 2026 г.

Methodology and Processing Parameters

This report presents the automated polarimetric analysis of dust continuum emission. The pipeline abandons arbitrary geometric cropping in favor of rigorous statistical isolation of dense molecular clumps.

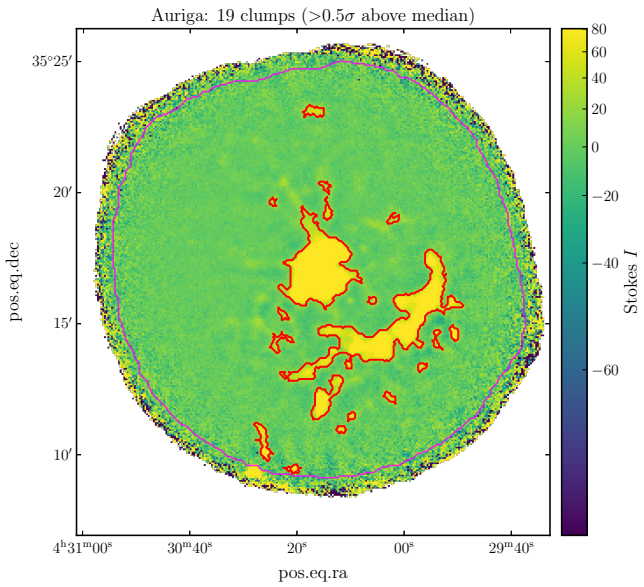
Statistical Segmentation and Masking: To mitigate edge artifacts, a geometric safety margin is established via distance transform. Within this zone, physical structures are isolated using a dynamic threshold based on the local median and standard deviation of Stokes I ($I > \tilde{I} + N\sigma_I$). Morphological dilation ensures the preservation of compact cores and the continuity of diffuse filaments. The background is strictly masked (assigned `NaN`), ensuring that polarization parameters and subsequent statistics are calculated exclusively within the identified physical structures.

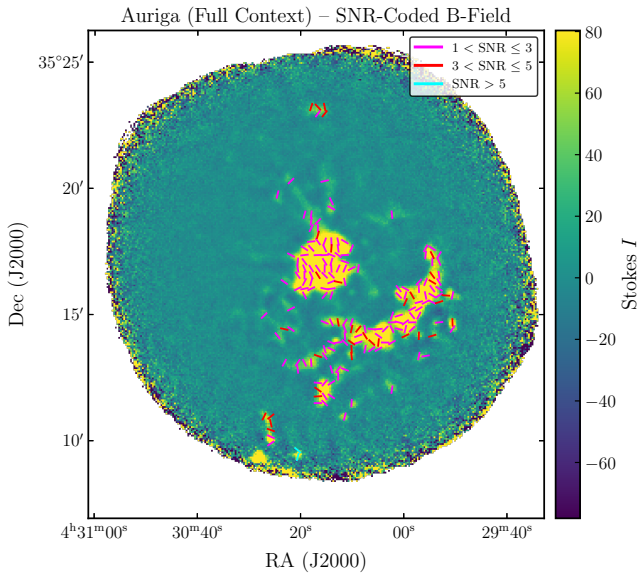
Plot Structure: For each targeted region, the analysis yields three diagnostic panels:

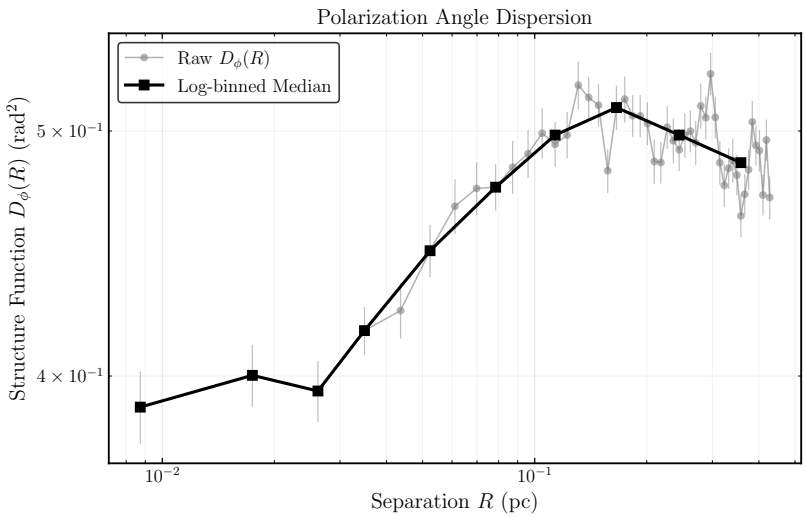
1. Statistical Segmentation (Top Left): Isophotes and morphological boundaries of the isolated dense cores over the asinh-stretched Stokes I continuum.
2. SNR-Coded Magnetic Field (Top Right): Polarization vectors (E-field rotated by 90°) overlaid on the full continuum map. Vectors are strictly constrained to the segmented areas and color-coded by the polarized intensity signal-to-noise ratio (SNR_{pI}).
3. Structure Function (Bottom Center): Angular dispersion of the magnetic field $D_\phi(R)$ plotted against the spatial lag R (in parsecs), calculated strictly within the isolated clumps.

Target: Auriga

Parameter	Value
Distance (pc)	450.0
Effective Pixels	4012
Stokes I RMS	42.87812
Median SNR	1.36
Distance Reference	Lada et al. 2009
Vector Decimation Step	4 px

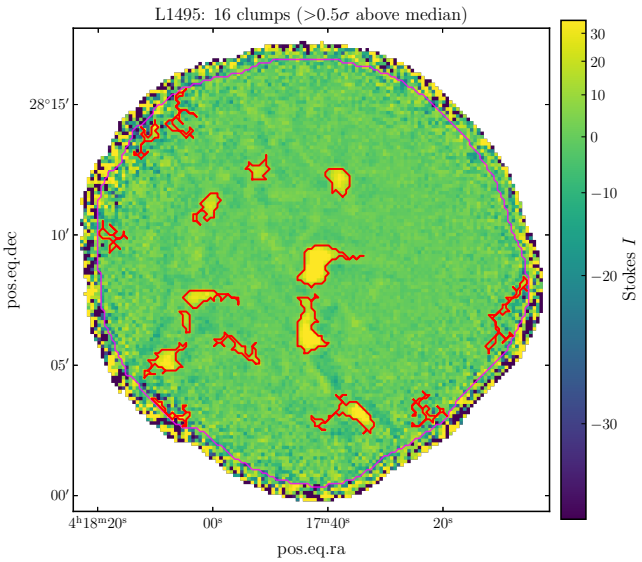


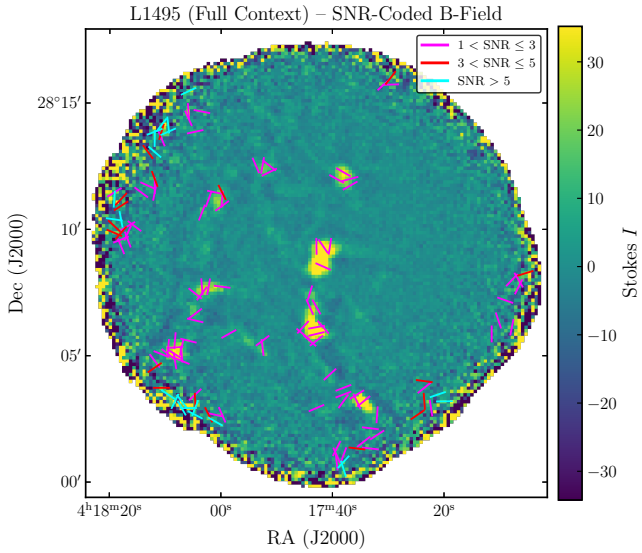


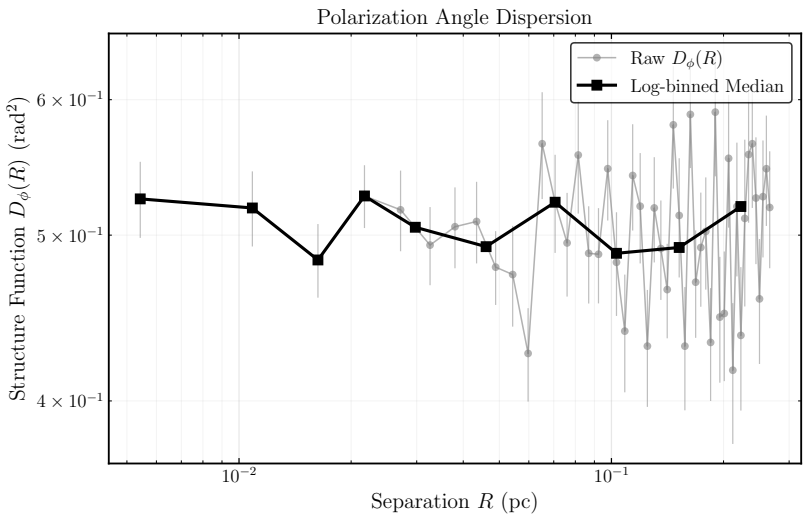


Target: L1495

Parameter	Value
Distance (pc)	140.0
Effective Pixels	953
Stokes I RMS	7.25939
Median SNR	1.46
Distance Reference	Galli et al. 2018 / Schlafly et al. 2014
Vector Decimation Step	2 px

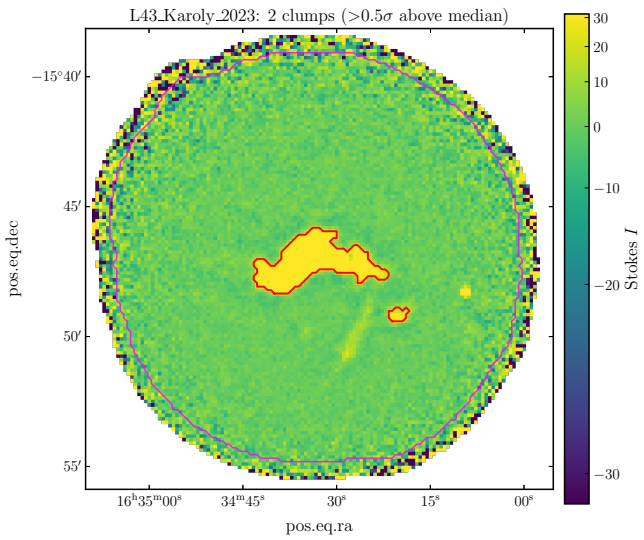


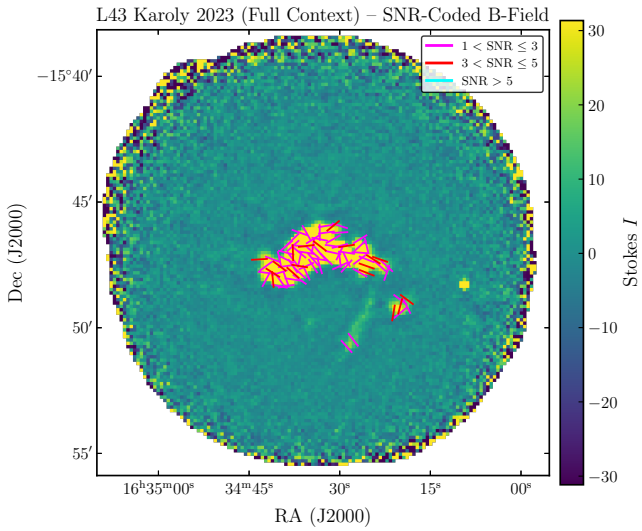


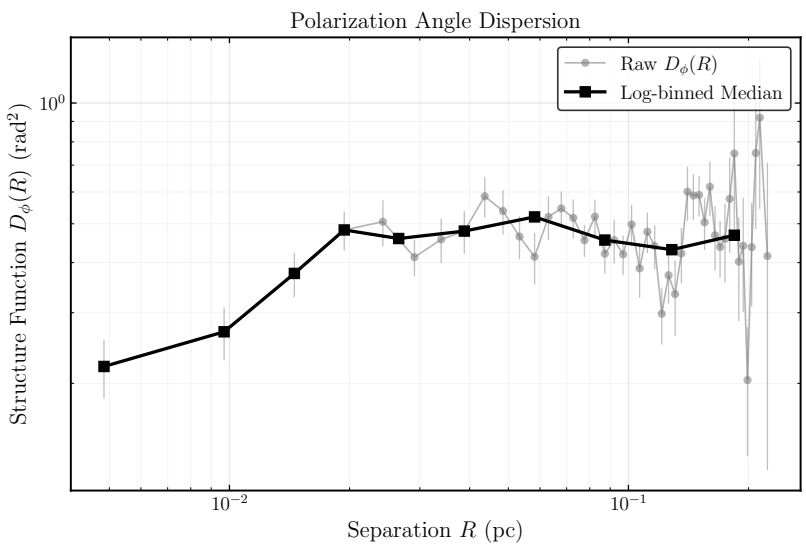


Target: L43_Karoly_2023

Parameter	Value
Distance (pc)	125.0
Effective Pixels	439
Stokes I RMS	48.02407
Median SNR	1.10
Distance Reference	Ortiz-León et al. 2017
Vector Decimation Step	2 px

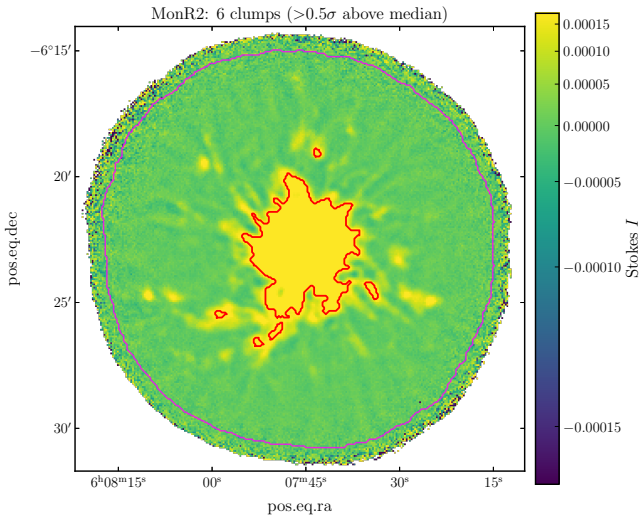


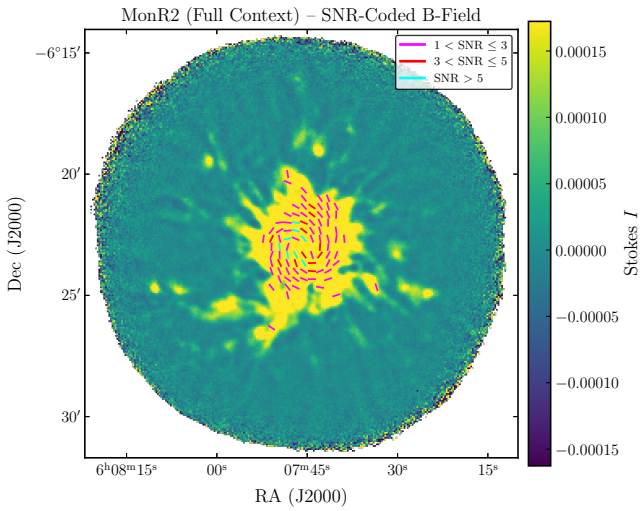


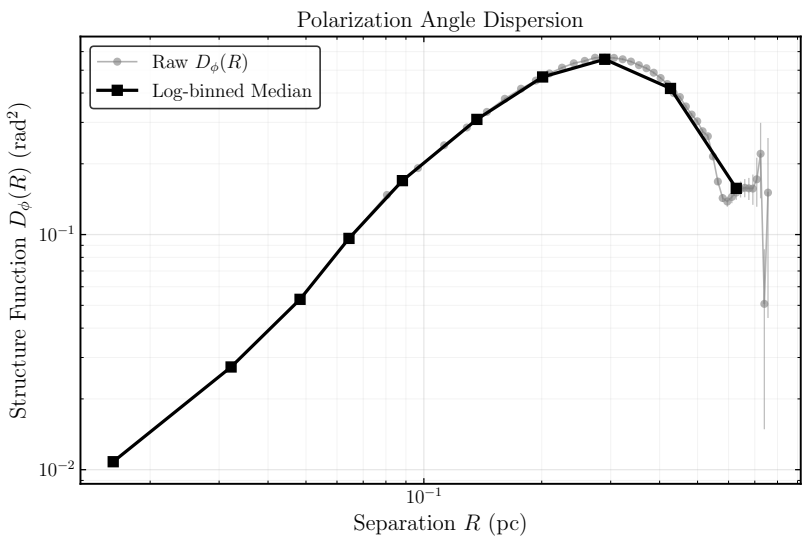


Target: MonR2

Parameter	Value
Distance (pc)	830.0
Effective Pixels	3671
Stokes I RMS	0.00042
Median SNR	1.33
Distance Reference	Dzib et al. 2016
Vector Decimation Step	5 px

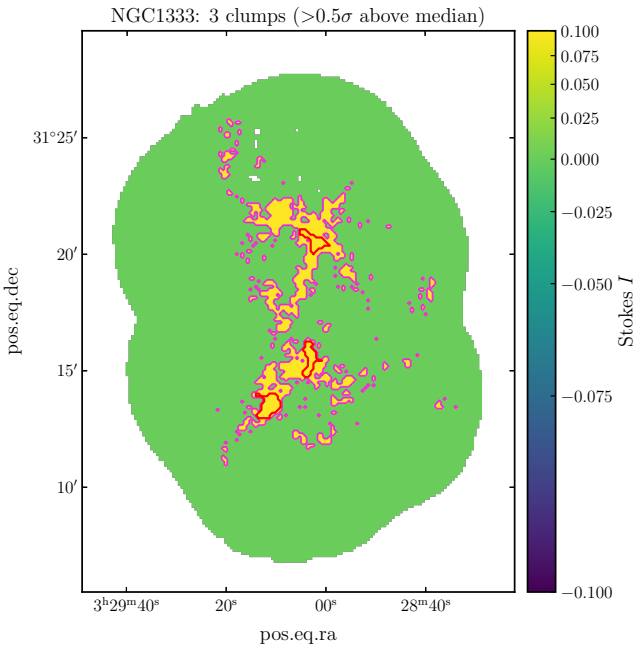


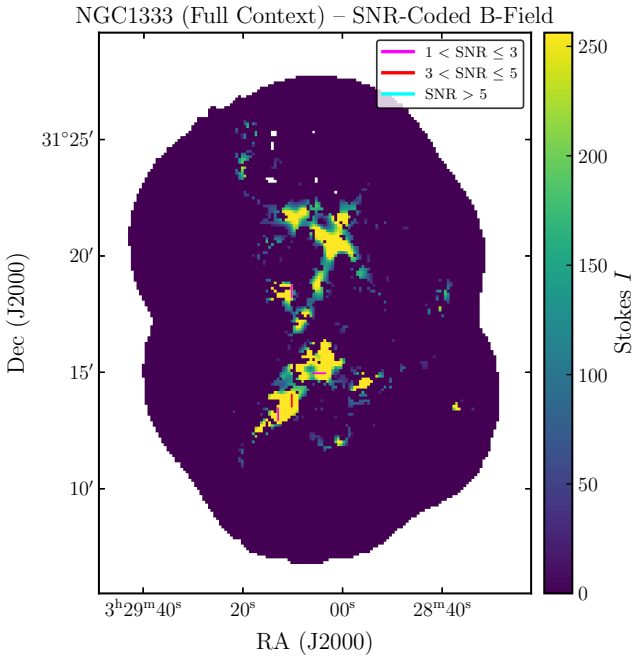


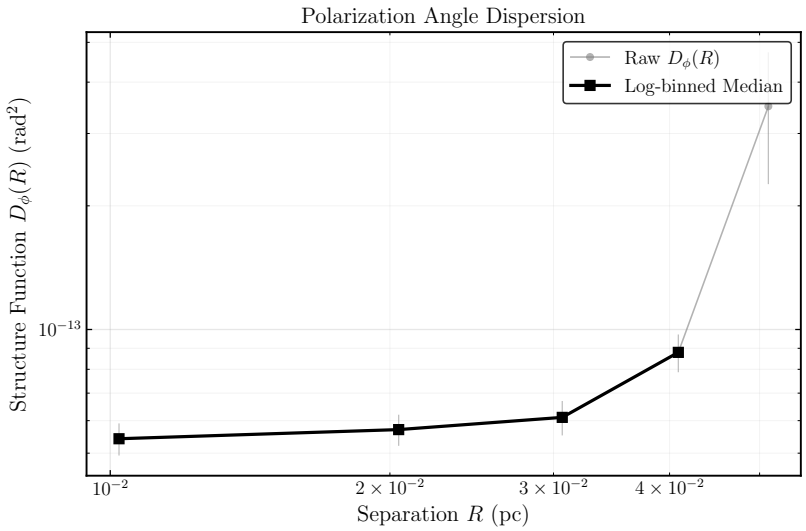


Target: NGC1333

Parameter	Value
Distance (pc)	299.0
Effective Pixels	202
Stokes I RMS	305.10187
Median SNR	2.00
Distance Reference	Zucker et al. 2018
Vector Decimation Step	5 px

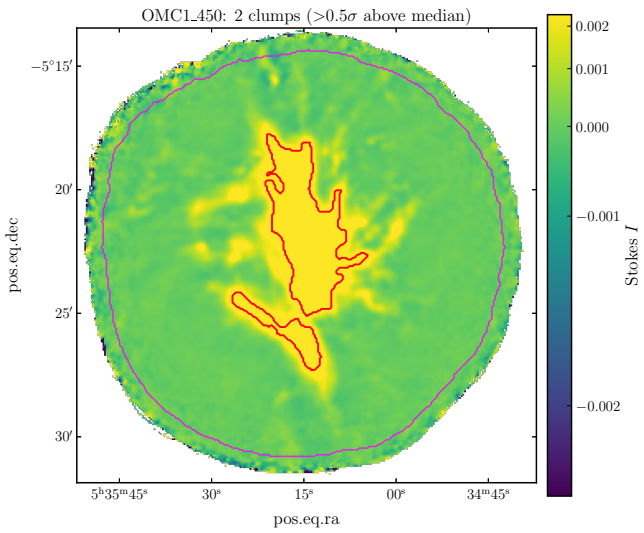


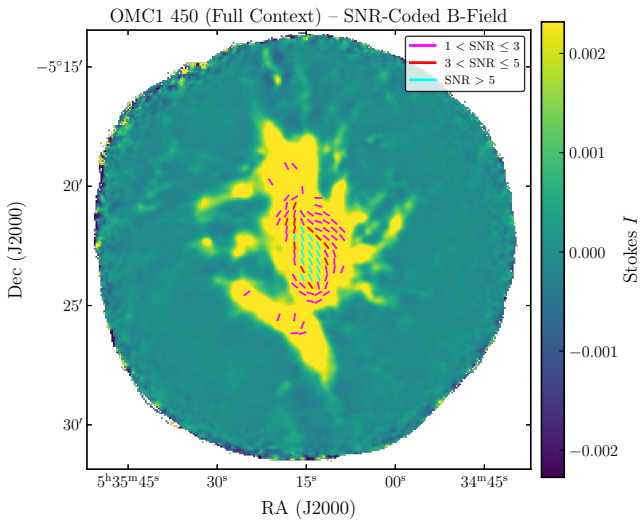


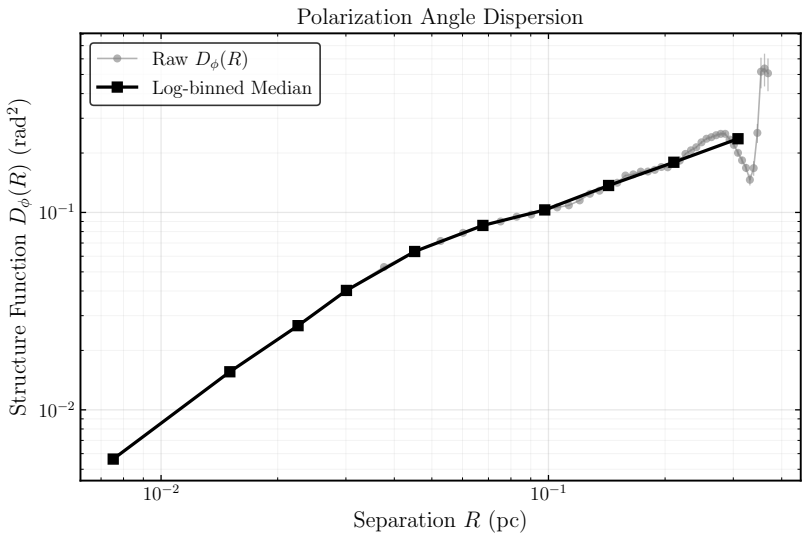


Target: OMC1_450

Parameter	Value
Distance (pc)	388.0
Effective Pixels	3841
Stokes I RMS	0.00537
Median SNR	1.64
Distance Reference	Kounkel et al. 2017
Vector Decimation Step	5 px

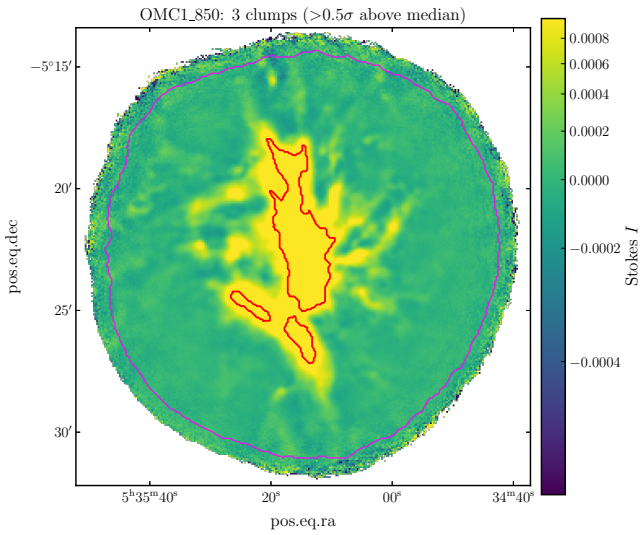


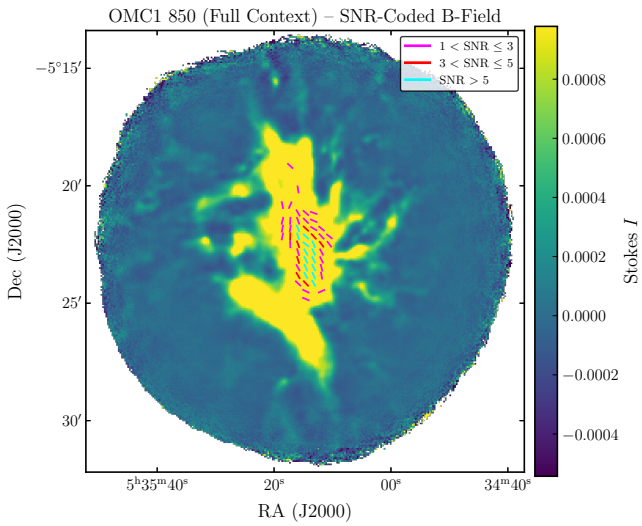


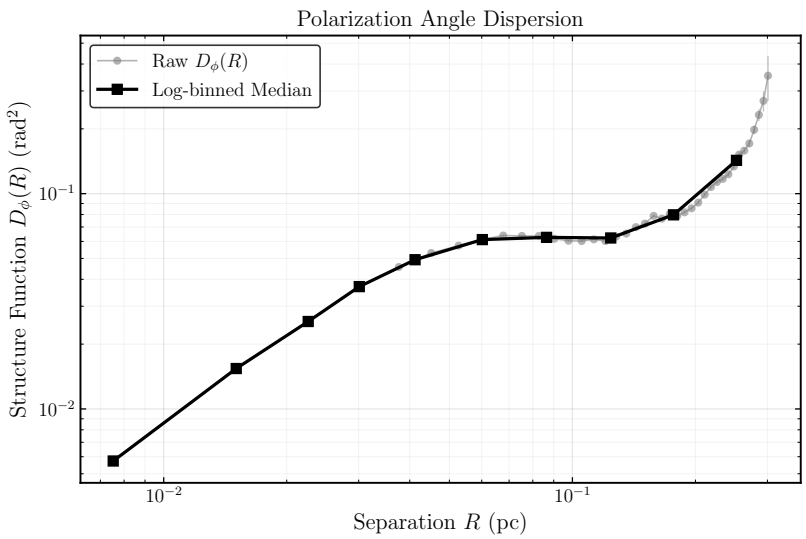


Target: OMC1_850

Parameter	Value
Distance (pc)	388.0
Effective Pixels	2670
Stokes I RMS	0.00226
Median SNR	1.72
Distance Reference	Kounkel et al. 2017
Vector Decimation Step	5 px

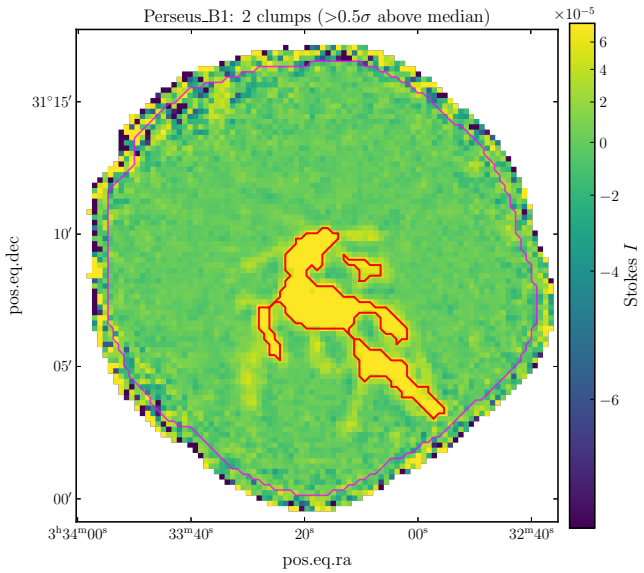


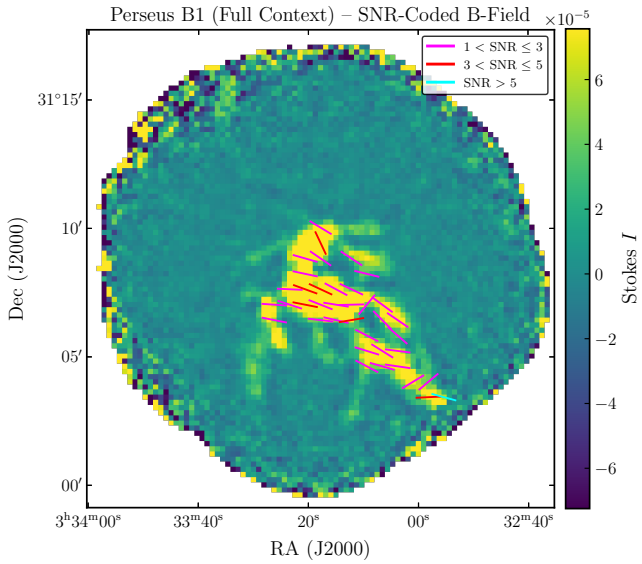


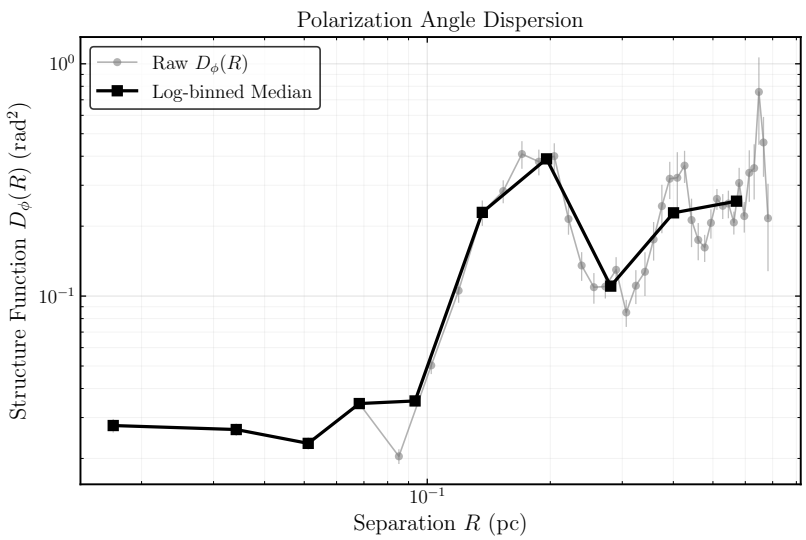


Target: Perseus_B1

Parameter	Value
Distance (pc)	293.0
Effective Pixels	385
Stokes I RMS	0.00006
Median SNR	1.49
Distance Reference	Zucker et al. 2018
Vector Decimation Step	3 px

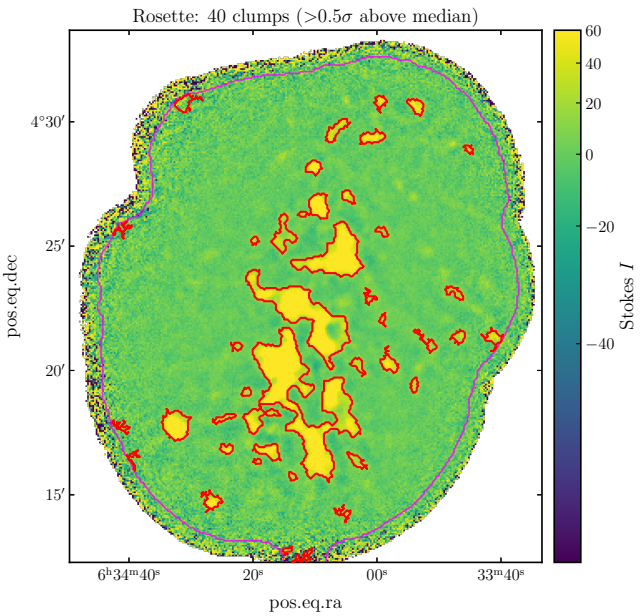


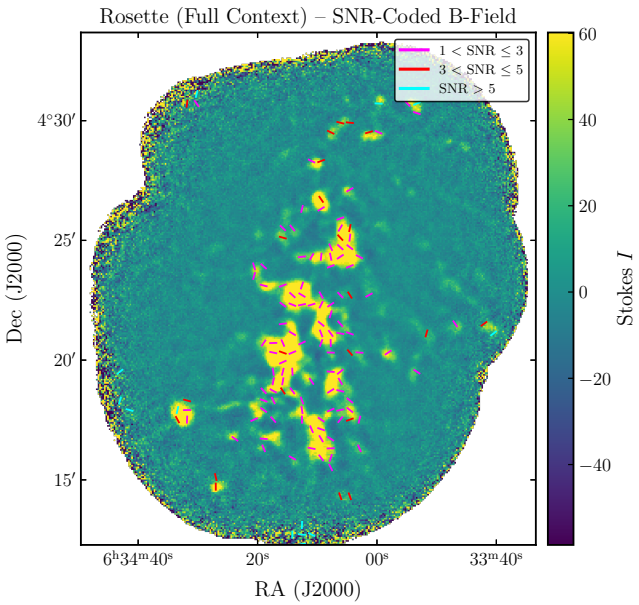


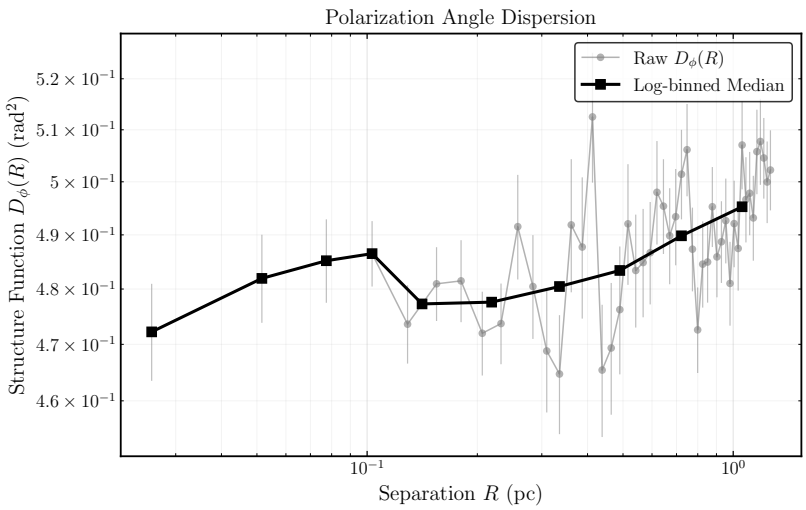


Target: Rosette

Parameter	Value
Distance (pc)	1330.0
Effective Pixels	6814
Stokes I RMS	25.34009
Median SNR	1.37
Distance Reference	Heyer et al. 2006
Vector Decimation Step	6 px

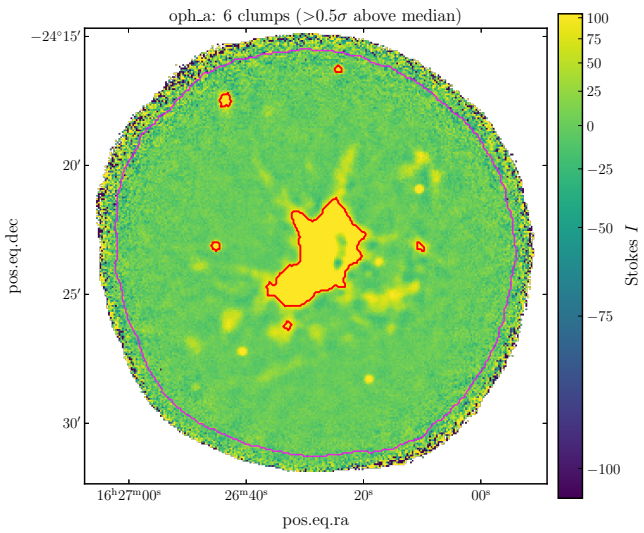


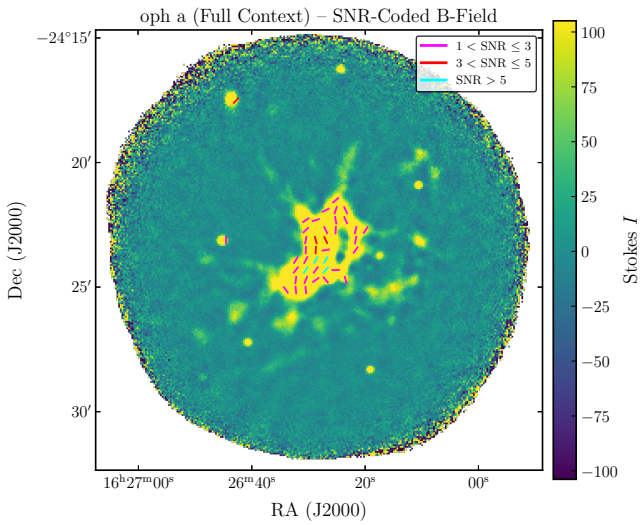


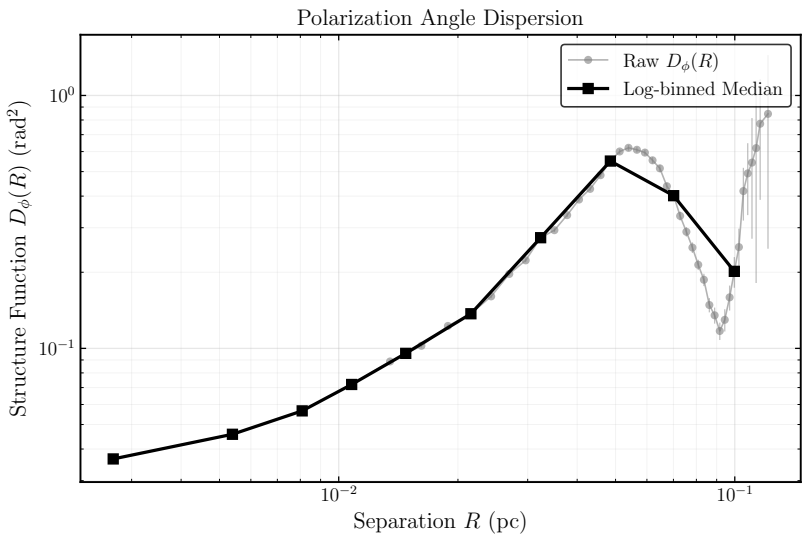


Target: oph_a

Parameter	Value
Distance (pc)	139.0
Effective Pixels	2135
Stokes I RMS	185.76798
Median SNR	1.27
Distance Reference	Zucker et al. 2019
Vector Decimation Step	6 px

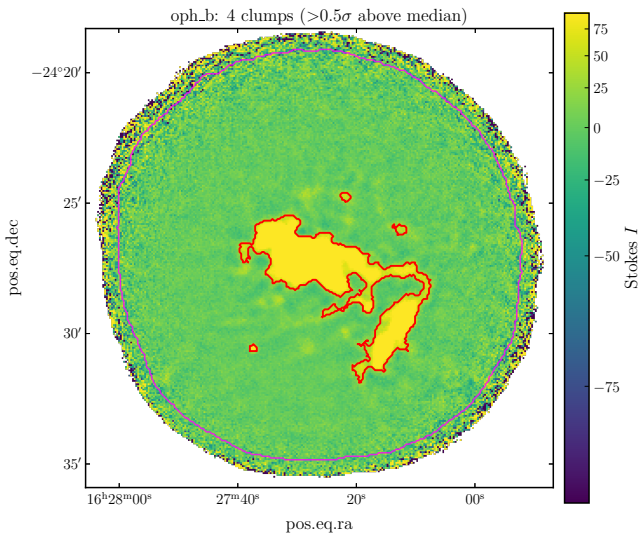


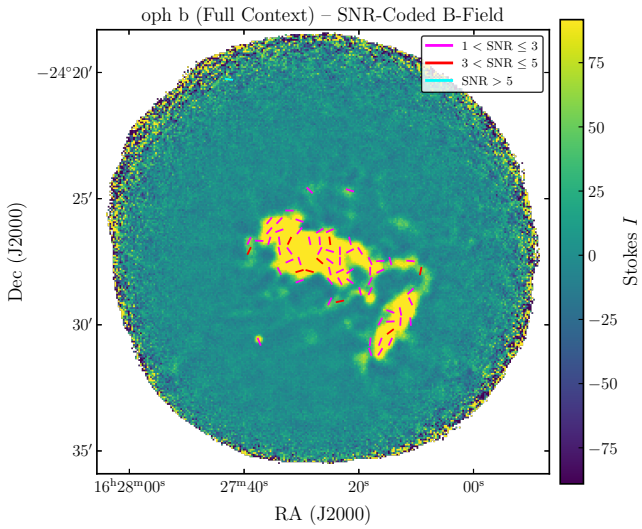


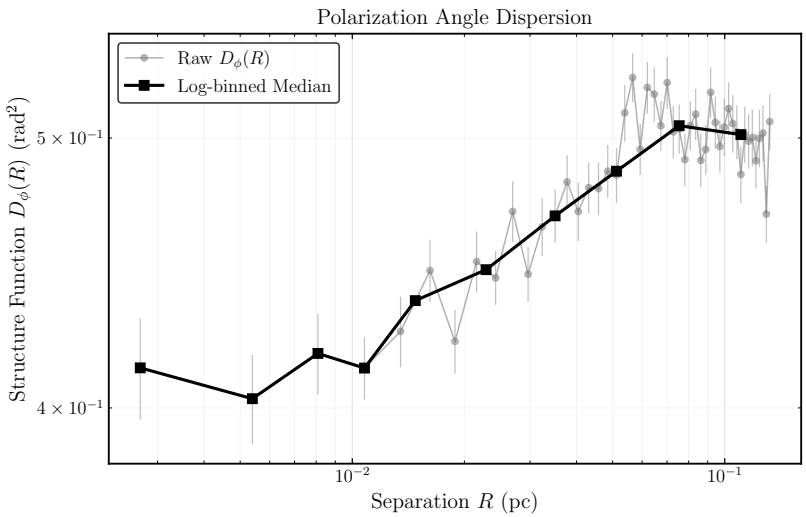


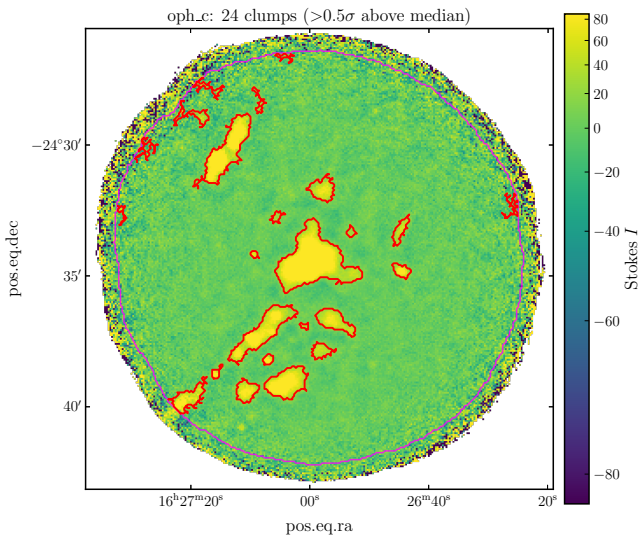
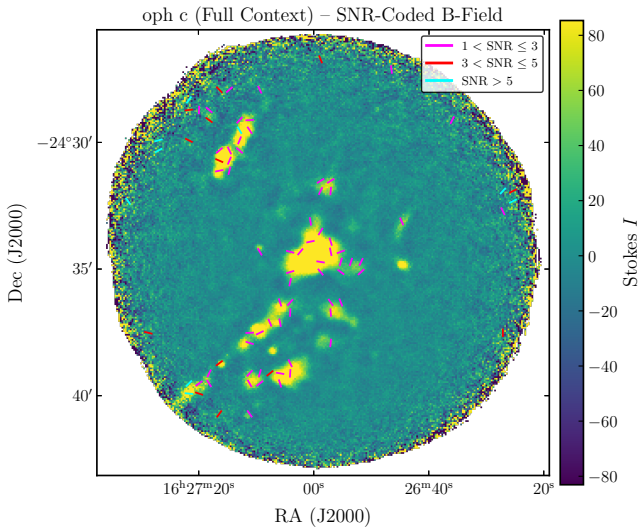
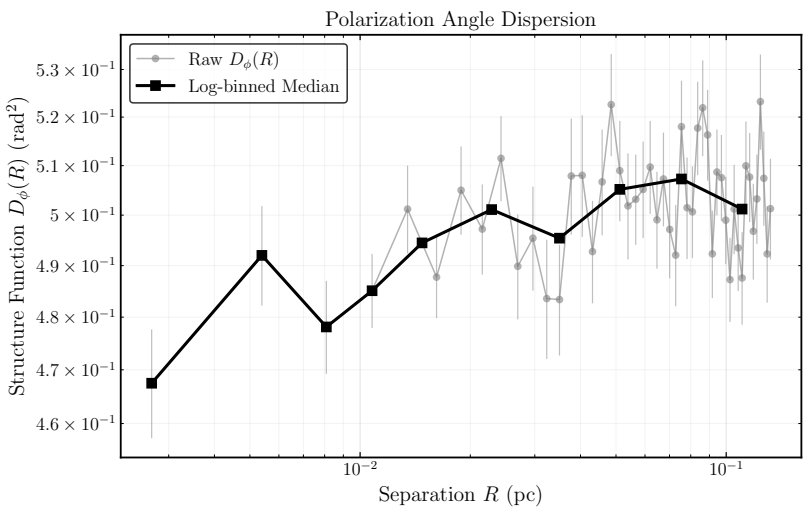
Target: oph_b

Parameter	Value
Distance (pc)	139.0
Effective Pixels	3503
Stokes I RMS	85.63193
Median SNR	1.16
Distance Reference	Zucker et al. 2019
Vector Decimation Step	6 px







Target: oph_c

Parameter	Value
Distance (pc)	139.0
Effective Pixels	4145
Stokes I RMS	33.63853
Median SNR	1.34
Distance Reference	Zucker et al. 2019
Vector Decimation Step	6 px